Engineering internship
Report

Darknet and CTI

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My huge thanks go to the team of Samapartners for making for me a welcoming and lovely environment to work and to integrate.

Abstract

The Dark Web, a conglomerate of services hidden from search engines and regular users, is used by cyber criminals to offer all kinds of illegal services and goods. Multiple Dark Web offerings are highly relevant for the cyber security domain in anticipating and preventing attacks, such as information about zero-day exploits, stolen datasets with login information, or botnets available for hire.

In this work, we analyze and discuss the challenges related to information gathering in the Dark Web for cyber security intelligence purposes. To facilitate information collection and the analysis of large amounts of unstructured data.

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I. Introduction

The Dark Web is a conglomerate of services hidden from search engines and regular Internet users. Anecdotally, it seems to the uneducated observer that anything that is illegal to sell (or discuss) is widely available in this corner of the Internet. Several studies have shown that its main content ranges from illegal pornography to drugs and weapons. Further work has revealed that there are many Dark Web offerings which are highly relevant for the cyber security domain. Sensitive information about zero-day exploits, stolen datasets with login information, or botnets available for hire can be used to anticipate, discover, or ideally prevent attacks on a wide range of targets. It is difficult to truly measure the size and activity of the Dark Web, as many websites are under pressure from law enforcement, service providers, or their competitors. Despite this, several web intelligences services have attempted to map the reachable part of the Dark Web in recent studies. One crawled the home pages of more than 6,600 sites (before any possible login requirement), finding clusters of Bitcoin scams and bank card fraud. Another study found that more than 87% of the sites measured did not link to other sites. This is very different from the open Internet, both conceptually and in spirit: in contrast, we can view the Dark Web as a collection of individual sites or separated islands.

II. Introduction of the host company

1) Company presentation

SAMA Partners is an IT & Engineering Consulting Company specialized in Cyber and Information Security, with Headquarter in Mannheim Germany, and offices in Munich (Germany) and Tunis (Tunisia).
It has high qualifications and many years of practical experience in various industries (e.g. finance, energy industry, transportation).
It has a Continuous organic growth since foundation in 2010 (Status 01.2020: 70 employees). it was Certified according to ISO/IEC 27001: 2013 (TÜV-SÜD) and Selected best midsized company of the year
2018 in Germany ("UnternehmerStar" 2018 Prize, Entrepreneurial Star Prize). It operates its own Cyber Security Academy offering several security courses and certifications.

SAMA Partners is a member of the German accredited competency network "Security made in Germany "and
it was the initiator and organizer of the yearly cybersecurity conference at Mannheim.

2) Consulting services

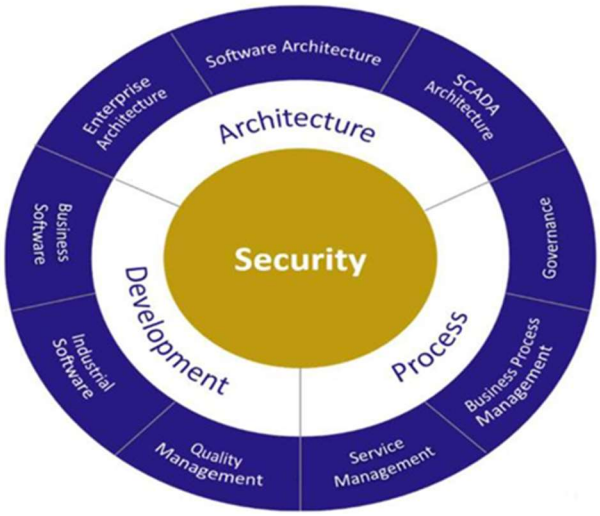


Figure 1 :SAMAPARTNER SERVICES

Security:

In SAMA Partners IT security is the top priority. Its consultancy team supports customers in the secure design of their corporate IT. From the use of relevant security technologies to designing and implementing a tailormade IT security architecture, in accordance with the law.

Architecture:

A secure enterprise and SCADA architecture provide a solid basis for enterprise IT. SAMA Partners supports its customers in building and optimizing their secure architecture and enables intelligent communication in their production using a customized platform for integration of Enterprise Applications.

Development:

The software used should always satisfy the exigences and map processes to optimum effect. Experienced experts of SamaPartners develop secure industrial and business software tailored to other companies. SamaPartners best practices and security standards are decisive for software development as well as for frontend and backend quality assurance.

Processes:

A lean and agile process management is needed to be efficient and to be able to react quickly to new requirements and customer wishes. With support from SAMA Partners, customers will receive not only secure business processes but also optimized service management for even more satisfied customers.

3) Costumers:

SAMA Partners has successfully established itself in the IT and engineering consulting sector. In addition to small and medium-sized customers, it serves several large companies in the region. Here are some customers:



Figure 2 : SamaPartners clients

4) Partners

Sama partners are:



Figure 3 :SamaPartners partners

III. Technical background

1) Surface Web

The Surface Web has been part of the World Wide Web since the first browser was first invented in 1990 by Tim Berner-Lee and is the part of the web you will be most familiar with, as it anything that can be discovered through your internet browser using any of the main search engines (Google, Bing, Yahoo etc.).

This is what you use when you read the news, buy something on Amazon, or visit any of your usual daily websites, and is also the area of the web that is under constant surveillance by governments across the world.

Making up under 5 percent of the total World Wide Web content that is known to exist is a figure that is
already miniscule, but estimates suggest that the Deep Web could be anywhere from 500 to 5000 times
larger. In perspective, there are roughly twenty terabytes (TB) of data and roughly one billion documents on

the surface web compared to 7,500 terabytes (TB) of discovered data and almost 600 billion discovered
documents on the deep web.

2) Deep web

The Deep Web was also part of the web at its conception, and in basic terms it is the opposite of the surface as it anything that search engines cannot find. This is the key difference between the two in real data terms. sites on the surface internet are indexed for search engines to find, but the deep web is not indexed.

However, both are also accessible by the public, they just require different methods to access them – usually a specific password encrypted browser or a set of logs in details.

Despite containing the Dark Web, it is not actually so bad and without it we wouldn't be able to use the

internet as we do today: The Deep Web contains all our medical records, financial records, social media files, and plenty other important information we want and need to keep secure. It is this need to keep secure files that gave rise to the need to keep a portion of the web secured away from being "Googled" at the whim of anybody at any time.

3) Dark web

The Dark Web IS part of the Deep Web, but it has one major difference in that it has been intentionally
hidden and is inaccessible to normal web browsers. The technology to create the Dark Web was initially
created (and still funded) by US Military Researchers in the mid-1990s to allow spies and intelligence agencies to anonymously send and receive messages. Named "The Onion Router", it was quickly coined with the shorter "Tor" with its name coming from application layer encryption within a communication protocol stack; many layers representing the layers of an onion.

If the US Military built it, why is it accessible to anyone with the right tools? The strategy was to release Tor into the public domain with simple logic; you cannot hide messages if there is nothing to hide them behind

therefore, if more people had access to send anonymous messages, the harder it would be for counterintelligence to discover their messages.

Another perceived benefit was to help people in nations where they are seen to be oppressed, with
impossible freedom of speech laws, to allow them to voice themselves freely where they cannot be tracked and punished.

A good idea in theory however it has primarily been filled with crime and the ability to find these criminals is extremely difficult - the whole point of the original Tor project was to be impossible to find.

Surface web	Deep web	Dark web
Accessible	Accessible by password, encryption, or through gateway software	Restricted to special browsers
Indexed for Search Engines	Not indexed for Search Engines	Not indexed for Search Engines
Little illegal activity	Little illegal activity outside of Dark Web	Large scale illegal activity
Relatively small	Huge in size and growing exponentially	Unmeasurable due to nature

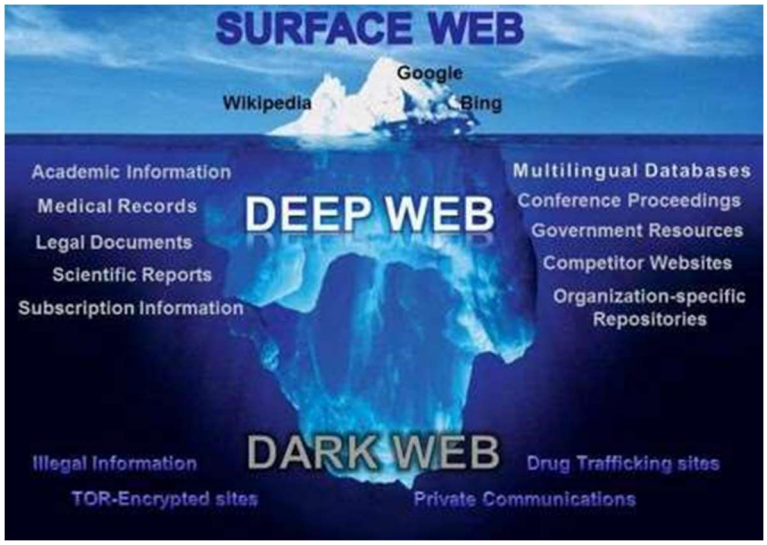


Figure 4 :Surface web vs Deep web vs dark web

4) Tor architecture

You may know Tor as the hometown of online illegal activities, a place where you can buy any drug you want, a place for all things illegal. Tor is much larger than what the media makes it out to be. According to Kings College much of Tor is legal.

Using Tor, your computer never communicates with the server directly. Tor creates a twisted path through 3 Tor nodes and sends the data via that circuit.

The core principle of Tor is onion routing which is a technique for anonymous & secure communication over a public network. In onion routing messages are encapsulated in several layers of encryption.

5) Tor hidden services

Applying the concepts of onion routing and telescoping, users obtain anonymity by sending their communication through a so-called Circuit of at least three relay nodes. As Tor is effectively a crowdsourced network, these relays are largely run by volunteers. The network has been an important tool for many Internet users who depend on anonymity, from dissidents to citizens in countries with restricted Internet access. Besides enabling users to connect to websites anonymously, Tor offers a feature called Hidden Services. Introduced in 2004, it adds anonymity not only to the client but also to the server, also known as responder anonymity. More concretely, by using such Hidden Services, the operator of any Internet service (such as an ordinary web page, including forums or message boards, which we are interested in for this work) can hide their IP address from the clients perusing the service.

When a client connects to the Hidden Service, all data is routed through a so-called Rendezvous Point. This point connects the separate anonymous Tor circuits from both the client and the true server. Figure 1 illustrates the concept: overall, there are five main components that are part of a Hidden Service connection. Besides the Hidden Service itself, the client, and the Rendezvous Point, it requires an Introduction Point and a Directory Server.

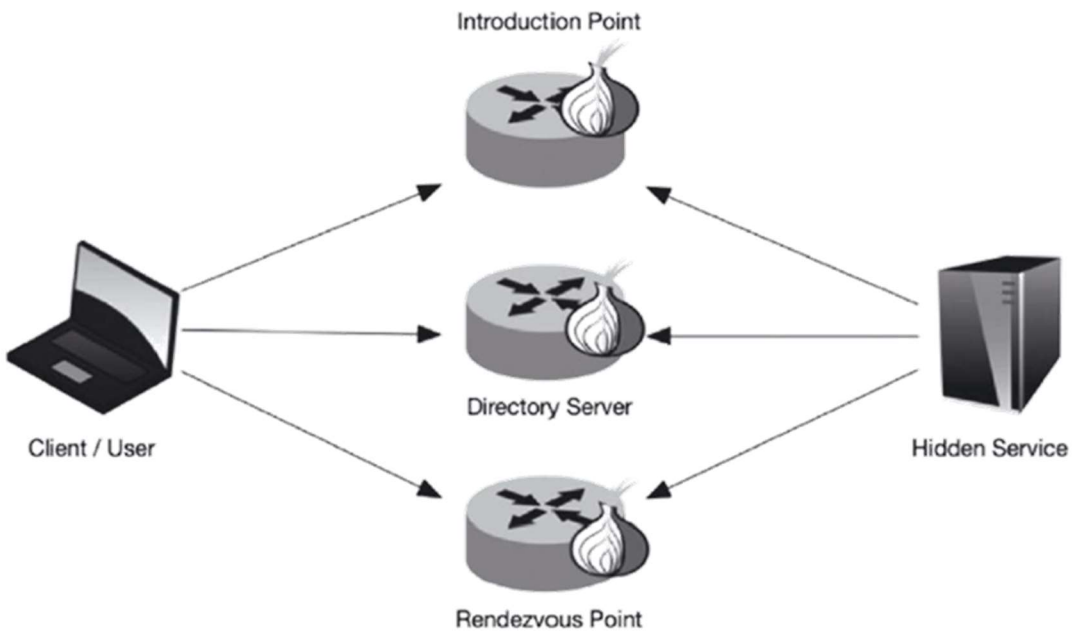


Figure 5 : hidden services architecture

The former are Tor relays, which forward management information necessary to establish the connection via the Rendezvous point and are selected by the Hidden Service itself, which is necessary to connect the client and the Hidden Service at the Rendezvous point. The latter are Tor relay nodes, where Hidden Services publish their information, and which are then communicated to clients to learn the addresses of the Hidden Service's introduction points. These directories are often published in static lists and are in principle used to find the addresses for the web forums.

It is unsurprising that Tor Hidden Services are a very attractive concept for all sorts of underground websites, such the infamous Silk Road or AlphaBay and due to their popularity form in effect the underlying architecture of the Dark Web.

IV. Challenges in dark web monitoring

The overarching main issues in analyzing the Dark Web for cyber security intelligence relate to the fact that a vast amount of unstructured and inaccessible information needs first to be found and then processed. This processing also needs to be done in a scalable way that enables humans to collect useful intelligence quickly and reliably.

In the following, I outline the concrete challenges that needed to be overcome in developing the solution for the dark web monitoring.

1) Acquisition of relevant target forum

The first challenge is the identification of target forums that are relevant to our operation, i.e., those that contain users and content relating to cyber security intelligence. Due to the underground nature of the intended targets, there is no curated list available that could be used as input to our algorithm. Intelliagg, a cyber threat intelligence company, recently attempted to map the Dark Web by crawling reachable sites over Tor. They found almost 30,000 websites; however, over half of them disappeared during their research [1], illustrating the difficulty of keeping the information about target forums up to date. Combined with the mentioned previously fact that 87% of Dark Web sites do not link to any other sites, we can deduce that the Dark Web is more a set of isolated short-lived silos than the classical Web, which has a clear and stable graph structure. Instead, only loose and often outdated collections of URLs (both from the surface Internet as well as Hidden Services) exist on the Dark Web. Consequently, a fully automated approach to overcome this issue is infeasible and a semi-manual approach must initially be employed.

2) Resources Requirement and scalability

Several technical characteristics of the acquired target forums require the use of more significant resource inputs. As is typical in analyzing large datasets obtained from the Dark Web, it is necessary to manage techniques which limit the speed and the method of access to the relevant data. Such techniques include the deliberate (e.g., artificial limiting of the number of requests to a web page) and the non-deliberate (e.g., using active web technologies such as NodeJS, which break the use of faster conventional data collection tools). Typically, these issues can be mitigated by expending additional resources. Using additional virtual machines, bandwidth, memory, virtual connections or computational power, we can improve the trade-off with the time required for efficient data collection. For example, by using several virtual private networks (VPNs) or Tor circuits, it is possible to parallelize the data collection in case there is a rate limit employed by the target. Surprisingly, a factor not challenging our resources was the habit of extensively vetting the credentials or 'bona fides' of forum participants before allowing access. A sufficient number of the largest online forums are available without this practice, which enabled data collection and analysis without having to manually circumvent such protection measures. However, since I did encounter at least some such forums (or parts of forums), our approach could naturally be extended to them, although this would require significant manual resource investment.

3) Globalized environment

As cyber security and cybercrime have long become a global issue, underground forums with relevant pieces of information are available in practically all languages with a significant number of speakers. Most existing studies of Dark Web content have focused on English or another single language. However, the ability to gather and combine information independent of the forum language broadens the scope and the scale of our solution significantly. By employing automated machine translation services, we are going to be able to not only increase the range of our analysis but also detect relationships and common threads and topics across linguistic barriers and country borders. Naturally, this approach comes with several downsides. For example, it is not possible to employ sentiment or linguistic analysis on the translated texts nor is the quality of state of-the-art machine translation comparable to the level of a human native speaker.

4) Real-Time intelligence extraction

Beyond the previous issues, our solution need to focuses in particular on the challenges posed by the nature of a real-time intelligence extraction process. Whereas previous studies have collected data from the Dark Web for analytical purposes, they have typically concentrated on a static environment. In contrast to collecting one or several snapshots of the target environment, our solution aims to provide intelligence and insights much faster. Real-time capability is a core requirement for the longer-term utility of the system, due to the often very limited lifetime of the target forums. To enable these functionalities, a high grade of automation is required, from the collection to the live analysis of the data. After the initial bootstrapping of sources and creating a working prototype, it is imperative that the processes require less manual input beyond normal human oversight tasks.

V. Architecture of the proposed solution

In this section, we describe the basic architecture of our solution. We largely abstract away from the exact technologies used and focus on the processing chain and the data model that enabled us to analyze the target forums in real time. Figure 2 shows the processing chain, including five phases defined as a recurrent cycle. The phases of the cycle are highly inspired by the conceptual model of the intelligence cycle. Like the intelligence cycle, theses phases are continuously iterated to produce new insights.

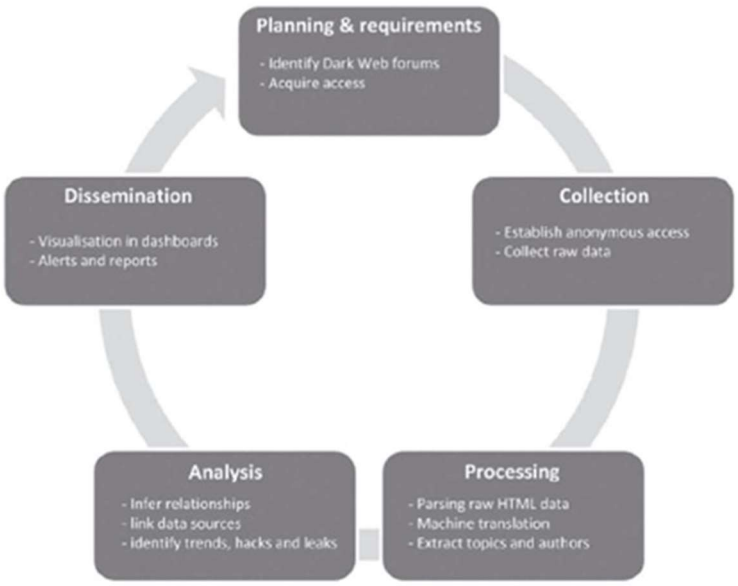


Figure 6 : process cycle of the solution

1) Planning and requirement

The key focus of our solution is on automation; manual work should only be needed for the integration of target forums in the initial planning and requirements phase, while all other phases are highly automated.

a. Identifying dark web forums

The first suitable target forums are identified by hand to bootstrap the process and overcome the challenges described in the previous Section. After obtaining a foothold, our solution then aims to analyze the content of these forums in order to obtain further links and addresses to other targets in a more automated fashion in later iterations.

b. Gaining access

Since most forums require some sort of login to access the site, our solution needs personal accounts to authenticate on each site. The way to acquire such logins differs on each site. While certain sites only request new users to provide a valid email address, others have higher entry barriers with reputation systems, measures of active participation, or even requiring users to first buy credits.

2) Collection

After the planning and requirements phase, all steps are fully automated. The collection phase deals with establishing anonymous access to the forums over Tor and the collection of raw data.

a. Establishing anonymous to forums

We establish anonymous gateways to the identified forums using Docker containers, Tor to access Hidden Services and Virtual Private Networks (VPN) for regular Deep Web sites. Here, it is necessary to add custom functions to our solution, which emulate typing and clicking behavior in order to log in automatically and subsequently detect whether the gateway has successfully logged into the target or not.

b. Collection of raw data

For the actual collection of the forum content and metadata, we employed the node.js headless Chrome browser puppeteer as a crawler within the Docker containers. While it requires more resources than other collection methods, it more closely emulates the behavior of real forum users, meaning it more easily avoids defensive action by the Dark Web marketplace operators. In order to improve the speed, the collection is distributed across multiple containers and parallelized.

3) Processing

The processing phase deals with parsing the collected raw HTML data from the previous phase, translating the content into English and extracting the entities of interest to feed a knowledge graph.

a. Parsing raw HTML data

Since our solution retrieves data over a headless browser, the data to process is in the Hypertext Markup Language (HTML). Extracting structured information from HTML data can be quite challenging depending on the layout of the forums. Our solution should implement a standard HTML parser that we adapt to the layout of each forum. While this approach may seem expensive at first, many forums have a similar layout such that the same parsers can be reused for different forums. The output of the

HTML parser for each page is a structured file including only the text information from the HTML page.

b. Translation of raw data in foreign languages

As much of the collected raw data contains content in several languages, we used automated machine translation to convert all non-English content into English. Through the use of Google's translation API, we obtain state-of-the-art translations, which enable the more complex data modeling and relationship analysis over forums in different languages in the follow-up phases.

c. Information extraction

To extract relevant information from the translated text from the gateways, we developed so-called extractors in Scala, which were also processed in a distributed fashion using the Apache Spark analytics framework. Our solution extracts information about the forum writers and their content, i.e. the titles of forum threads and the posted messages. It then constructs a knowledge graph that connects threads, actors, messages, and topics. Figure 7 shows the underlying data model of the knowledge graph of our solution. The collected raw data and the knowledge graph is then put into Elasticsearch, a search engine based on Lucene. As a tool for data exploration, it reads structured data and interprets timestamps and locations.

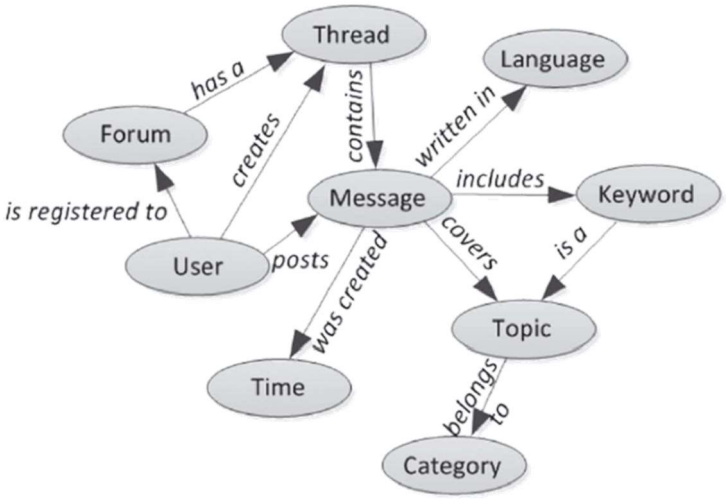


Figure 7:data model representing the knowledge graph

4) Analysis

While inferring simple relationships between messages and authors is a relatively easy task given the HTML structure of the forums, other types of relationships and information extraction steps for the knowledge graph require advanced data analysis techniques. our solution's goal is to automatically find relationships and trends across

different threads and forums; the following processing steps are thus executed in this phase.

a. Infer user relationship

Relationships between users are mainly inferred in our solution through the analysis of threads since users of Dark Web forums barely link to each other explicitly as in classical social networks such as Facebook or LinkedIn. A thread is always created by a single user and many different users then start posting messages on this thread. Our solution infers a relationship between users by ordering all messages in the same thread by their message times. We define a relationship from user A to user B if user B posted a message after user A in the same thread. The intuition is that user B is interacting with user A when he replies to his messages.

b. Identify cyber security trends

To identify cyber security trends, our solution fuses the messages, topics and categories from the different forums and computes aggregated time series. These time series form the basis to identify trending topics, e.g., when the time series experiences a high growth or decline over short periods. Long-term trends are also detectable given that all collected messages are time-stamped and thus provide information over the whole lifetime of the forum.

5) Dissemination:

Finally, it is important to disseminate the extracted information so that it can be easily processed by human intelligence analysts. To serve this purpose, our solution should support various types of data visualizations and data query interfaces for exploratory analysis. For example, customized Kibana dashboards provide realtime views of the processed data that is stored in the Elasticsearch database. These dashboards can be generated and customized easily by the users allowing different views depending on the question of interest.
Finally, users may realize that some data is missing or that the additional forums should be integrated. The cycle of our solution's architecture supports users to refine the planning and data collection requirements, thus closing the loop of the intelligence process.

Conclusion

It is imperative in the current cyber security environment to have a real-time monitoring solution that works across languages and other barriers. We have shown in this paper that early detection of cyber threats and trends are feasible by overcoming several key challenges towards a comprehensive framework. While we can be certain that techniques like ours are being used by both governmental and private intelligence actors around the world, it is important to analyze their power in a more open fashion, giving rise to possible scrutiny and further development